Problem: Implications and Quantifiers

Let the domain of p be all network packets captured by an interface, and the domain of s be all known malicious signetures.

- Let $M(p,s)$ denote: "packet p matches signeture s."
- Let $D(p)$ denote: "packet p is dropped by the firewall."

1- Translate the following statement into formal logic:

"Eevery packet that matches ~~the~~ at least on malicious signature is dropped by the firewall"

2- Find the logical negation of your expression in (1), push the negation operator inward until it directly applies to the predicates

3- Rewrite the resulting negated expression in clear English

Solution:

1- $\forall p\,((\exists s\, M(p,s)) \rightarrow D(p))$

2- $\neg(\forall p\,((\exists s\, M(p,s)) \rightarrow D(p)))$

$\exists p\,((\exists s\, M(p,s)) \wedge \neg D(p))$

3- "there is at least one packet that matches a milicious signature, but it dose not dropped by the firewall"